

Review

In Silico strategies for drug discovery: optimizing natural compounds from foods for therapeutic applications

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Abstract

The physicochemical properties of natural chemical compounds serve as a foundation for the development of novel drugs and innovative therapies. While several databases describe properties of natural products, their applicability and data accessibility are limited. Thus, the lack of accessible data represents a significant challenge in developing drugs based on natural compounds. Although chemical properties of natural compounds can be determined experimentally, this approach requires costly materials and procedures. In silico alternatives for drug analysis and pharmaceutical design cycles represent an interesting, simpler, and less expensive option for natural compound-based drug discovery. This article examines in silico methods for the characterization, design, and optimization of natural compound-based drugs derived from food. The review focuses on how in silico-based tools, such as machine learning, computer-based mathematical modeling, homology prediction, docking, molecular dynamics, and simulated molecular evolution events, are used to optimize natural compound testing and design. The in silico bioactivity predicted properties for peptides and secondary metabolites are discussed. In silico analysis is also explored as a tool to predict the antioxidant, antidiabetic, antimicrobial, and cardiovascular effects of natural compounds from foods. The approaches here presented can help speed up the discovery and development of natural compound-based drugs for therapeutic use.

Keywords In silico · Natural compounds in foods · Homology prediction · Simulated molecular evolution events · Bioactivity · Secondary metabolites

1 Introduction

Since ancient times, human civilizations have obtained natural products to satisfy basic needs such as food and shelter [1]. Natural compounds have attracted the attention of diverse communities as base components for traditional and herbal medicines [2]. Even today, natural substances obtained from plants, marine life, fungi, and bacteria are still a vast source of inspiration for modern drug discovery [3, 4]. Approximately two-thirds of modern small molecule drugs approved by drug administration agencies are somehow related to natural compounds [2, 5]. This enduring relevance highlights the potential of natural compounds as a critical foundation for discovering novel therapeutic agents in the ever-evolving field of drug development.

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Understanding the physicochemical properties of natural compounds serves as a starting point for new drug synthesis and design [2, 7, 8]. Even though properties of natural products are known and/or reported in diverse databases, most of the information available regarding content, coverage, and relevance for natural products is still limited [7]. In 2020 for instance, around 250 k natural compounds were available in chemical datasets while the properties of only 25 k were listed as available from commercial sources and research institutes [2]. The limitations on data availability represent a challenge for the design of new drugs based on natural compounds. Beyond limited information, sourcing, extracting, and testing bioactive compounds from natural products remain key obstacles in developing new drugs [2]. The lack of accessibility of natural compounds' properties further confirms how experimental evaluation of chemical attributes represents a bottleneck in natural compound-based research.

In silico methods encompass a variety of computational algorithmic techniques used to predict physicochemical properties of chemical compounds. These methods serve as powerful alternatives in drug design by enabling rapid and high-performance predictions across multiple chemical property datasets. In silico-based analysis methodologies do not require isolation of chemical compounds, allowing the determination of health benefits before synthesis or purification. Modern desktop computers have sufficient processing power to run simulations of low to moderate complexity, making in silico research methods more accessible. However, these methods still require specialized computational skills and expertise, which may not be widely available. Computer-based in silico alternatives thus serve as cheaper alternatives to assist in research of drug analysis and design cycles. Laboratory work and material costs are therefore reduced through in silico analysis [6]. Although labor-intensive and time-consuming, lab experimentation remains essential, as in vitro or in vivo assays are still crucial for validation of in silico results.

This review explores the application of various machine learning-based techniques, homology prediction, molecular docking, molecular dynamics simulations, and simulated molecular evolution events in natural compound-based research. It highlights how these advanced methods have been employed to enhance the characterization, optimization, and design of bioactive natural compounds, providing deeper insights into their therapeutic potential and mechanisms of action.

2 In silico methods: prediction, modeling, and operation

Mathematical modeling refers to the creation of an abstract representation of a real-world scenario. A math model is designed to predict certain characteristics, or to provide insights into particular aspects of the modeled phenomena. In natural compound research, modeling methods have been used to successfully identify bioactive natural compounds. For instance, methods based on graphical analysis of 3D structures include molecular shape similarity analysis, pharmacophore models, molecular docking, and machine learning approaches for the virtual screening of bioactive natural compounds [9]. In this context, given the immense diversity of natural compounds and the lack of available structural data, the docking properties of natural compounds to macromolecules are rarely accessible. Therefore, homology modeling methods are an effective strategy for predicting docking and assessing molecular dynamic simulations in natural compounds [2].

Several studies have reported applying in silico approaches to identify natural compounds with potential health benefits and predict allergenic effects. Rost et al. [10] demonstrated, through in silico allergen analysis, how macadamia nuts contain proteins with allergenic and cross-reactivity potential. Computational methods have also been applied to simulate the release bioactive peptides using open-access tools. BIOPEP-UWM (<https://biochemia.uwm.edu.pl/bio-pep-uwm/>) is an open-access program designed for identifying and characterizing bioactive peptides. Similarly, ExPASy (<https://www.expasy.org/>) provides tools for simulating the digestion of known proteins, and for conducting proteomic sequence and structure analysis. ExPASy utilizes sequences from databases such as PDB (<https://www.rcsb.org/>) and PDBe (<https://www.ebi.ac.uk/pdbe/>) [11].

In silico methods provide tools to understand the interactions between biological molecules at a systemic level using network pharmacology analysis. This approach offers insights into the mechanisms of action of drugs, identifies new drug targets, predicts side effects, and uncovers potential drug-drug interactions [12]. Biological networks are often constructed based on experimental data or computational predictions and can represent different types of interactions, such as protein–protein interactions, drug–target interactions, or gene regulatory networks. Leveraging in silico approaches can enhance the bioactive effect of natural compounds. For instance, the Luna-Suárez group identified hypervariable regions on 11S globulin from amaranth were modified by inserting antihypertensive peptides (VY), resulting in a protein with a higher antihypertensive effect (8 to tenfold) when compared to the wild-type 11S globulin protein [13, 14].

It is well known that different compounds isolated from foods can exhibit bioactive properties or confer health benefits. Computational approaches offer strategies to corroborate these effects with greater accuracy. In this context, different methods can be used to model and predict the mechanisms of action of natural compounds, helping to elucidate their potential benefits and functional pathways.

2.1 Machine learning and modeling

Machine learning is a field of computer science that focuses on developing algorithms and statistical models that allow computer systems to learn and discern patterns. In simple terms, machine learning enables computers to recognize trends and make decisions based on data, much like how humans learn from experience. Machine learning-based systems attempt to mimic complex (human) learning processes by learning from experience. Instead of following rigid, pre-programmed rules, these systems improve over time by analyzing large sets of information. Such systems are trained to recognize patterns and make predictions without being explicitly programmed to do so based on experience gathered from large amounts of data [15]. There are numerous applications of machine learning-based systems; some examples are listed below.

- For natural compound analysis, machine learning-based analysis has been primarily used to predict chemical properties based on the structural characteristics of different compounds. This means that AI models can determine how a compound might behave based on its molecular shape and composition.
- The non-supervised natural language-inspired techniques are used to describe individual molecular structures in a database as elements in vector spaces to predict physicochemical properties [16]. In other words, these methods translate molecular structures into mathematical representations, allowing computers to compare and predict their physical and chemical behaviors.
- Virtual Screening can describe bioactivities for natural compound ligands that act as hormone receptor modulators in breast cancer-related research [17]. This helps us to quickly identify potential drug candidates for diseases like breast cancer.
- Molecular fingerprints are used to predict, *in silico*, the toxicity and accessibility properties of natural compounds in pharmaceutical drug development [18]. These digital "fingerprints" help to assess whether a compound is safe and effective before testing it in the lab.
- The *in silico* analysis of the physicochemical properties of sweeteners is used to predict the structural properties responsible for certain organoleptic properties of natural sweeteners [19, 20]. This means AI can help determine why some sweeteners taste sweeter than others by analyzing their molecular structures.

2.2 Homology prediction

Homology prediction, also called protein homology modeling, employs computational methods to predict unknown three-dimensional protein structures based on amino acid sequences. Proteins with similar amino acid sequences are likely to have similar structures. Thus, homology prediction compares the sequence of a target protein to known sequences of similar protein structures to predict the target's shape properties [21]. This approach homology prediction enables the rapid generation of structural predictions, allowing researchers to understand proteins to answer pharmaceutically relevant questions [22]. While computational structure prediction methods provide cost-effective alternatives to predict protein shapes when no experimental data is available, developing accurate models remains a challenge [23].

Homology modeling techniques have many applications in natural compound research. Homology prediction has been used to model G-protein-coupled receptors of fatty acid receptors involved in colorectal carcinoma. In particular, by understanding the structural properties of the G-protein coupled receptor (GPR120), natural products such as silibinin, withanolide, limonene, and curcumin have been shown to interact with the receptor. The interactions between these natural compounds and the receptor suggest their potential as anti-colorectal cancer therapeutics [24]. In another study, acteoside, a non-peptide natural compound, was found to inhibit inflammatory responses by antagonizing the inflammatory mediator-coupled receptor protein C5aR [25]. In addition, natural compounds have been reported to display antiviral properties. *In silico* methods such as Molecular Dynamic Simulators and Homology prediction have been used to predict the antiviral activities of natural compounds for designing new drugs to treat SARS-CoV-2 [26].

2.3 Docking and molecular dynamics

Docking is a computational approach that identifies potential molecules with biological activity by understanding how they bind to proteins or enzymes with important biological functions. Docking techniques thus helps us to understand how molecules might interact with key proteins in the body, a crucial step for new drugs development and for understanding biological processes.

Molecular docking simulations are done through data recovered from protein or genomic databases, which are used to identify the most favorable binding arrangement between the ligand (natural compound) to the target (key enzyme). Large amounts of data are therefore required to predict how well a particular molecule can attach to a specific protein, much like finding the perfect key for a lock.

Molecular docking and molecular dynamics are employed to screen and optimize natural compounds structures and to predict interactions between molecules. Molecular dynamics is commonly used to study the intermolecular interactions at the atomic level and the structural dynamic behavior of macromolecules [27, 28]. In other words, molecular dynamics helps us to understand how molecules move and interact with each other over time. These interactions include forces like electrical attraction (electrostatic interactions), weak bonding (hydrogen bonds and van der Waals interactions), and water-repelling effects (hydrophobic interactions), all of which influence how stable a molecule-protein interaction is.

As a result of molecular docking simulations, different scores are obtained to rank the best binding mode and interaction and, consequently, identify the most convenient bioactive compound. These scores help us determine which molecules have the strongest and most stable connection with the target protein.

Usually, the scores considered are: the binding energy score, which is a measure of the strength of interaction between the ligand and its receptor, typically reported in kilocalories per mole (kcal/mol), the lower the binding energy score found, is the better the protein–ligand binding stability [29]; the root-mean-square deviation (RMSD), that is used to measure the deviation in conformation that occurs throughout a molecular dynamic simulation with a reference structure; the solvent accessible surface area (SASA), and the radius of gyration (Rg), both measurements allow to evaluate physicochemical properties, as hydrophobic or hydrophilic regions exposed in the surface of ligand and its size and compactness, respectively. SASA and Rg are used to provide information about the thermodynamic properties of the system ligand-target studied, as described by Zhao et al. [11].

Table 1 shows different studies and briefly describes the strategies followed through an *in silico* approach to demonstrate the bioactive potential of different natural compounds through Docking and molecular dynamics simulations. Natural compounds can be seen as peptides, volatile compounds, polyphenols, and others from vegetal sources such as seeds, stems, roots, fruits, and leaves. Also included are some compounds, such as peptides and proteins, from animal sources such as milk, egg, meat, and insects. Table 1 lists various *in silico* studies using docking and dynamics molecular to assess the bioactive potential of natural compounds from diverse sources. Antidiabetic peptides from black beans showed stronger DPP-IV inhibition than sitagliptin, while betahistine from *Amaranthus* species demonstrated anti-obesity effects via lipase inhibition. Endophytic fungi-derived polyphenols exhibited antioxidant properties. Antimicrobial peptides from macadamia and ACE-inhibitory peptides from fish byproducts were identified as promising for hypertension treatment. Additionally, camel milk peptides and edible cricket peptides showed cholesterol-lowering and ACE-inhibitory effects, respectively, with binding affinities comparable to known drugs. These studies highlight *in silico* approaches as efficient tools for predicting bioactive potential in natural compounds.

2.4 Simulated molecular evolution events

During simulated molecular evolution events, the sequence composition of cellular molecules (such as DNA, RNA, and proteins) undergoes changes across generations. Simulating molecular evolution events, combined with *in silico* optimization, offers an alternative approach for generating genetic variations in populations of compounds, such as proteins or small molecules (phytochemicals), over time. This method lends itself to practical use through iterative design–synthesize–test cycles, where composition sequences are continuously optimized toward a specific computational or experimental objective. During each cycle, different compounds with certain targeted variations are used to induce changes (such as modifications in the amino acid sequence of a protein or changes in the chemical structure of small molecules) that lead to identifying new natural compounds with improved bioactivity [35, 36].

Table 1 In silico studies on natural compounds to demonstrate bioactive potential with Docking and Molecular Dynamics strategies

Source	Compound/s of interest	In silico approach	Outcome	Reference
Black bean (<i>Phaseolus vulgaris</i> L.)	Antidiabetic peptides (EGLELLLLLAG, AKSPLF, FEELN, TTGGKGGK, AKSPLF, WEVM)	Molecular docking analysis was performed to predict individual peptide biological potential using Docking-Server®	Peptides EGLELLLLLAG, AKSPLF, and FEELN inhibited DPP-IV more efficiently in silico through free energy interactions than the control sitagliptin	[30]
<i>Amaranthus aspera</i> L., <i>Amaranthus viridis</i> , and a leafy vegetable <i>Amaranthus tricolor</i> L	Betahistine, α -tocopherol, γ -tocopherol, phytanadione, and tocopheryl acetate	Interactions between Pancreatin lipase and specialized bioactive components and antioxidant-rich extracts in lipolysis regulation were checked by in silico docking	Betahistine showed higher antiobesity properties by lipase inhibition – 4.39 kcal/mol	[31]
<i>A. amstelodami</i> from <i>O. basilicum</i> , <i>A. niger</i> from <i>A. ampeloprasum</i> and <i>N. sativa</i> , <i>A. versicolor</i> from <i>P. nigrum</i> , <i>C. madrasense</i> from <i>P. oleracea</i> , <i>P. chrysogenum</i> from <i>A. cepa</i> , <i>P. maritimum</i> and <i>P. somniferum</i> , <i>P. citrinum</i> from <i>A. retroflexus</i> , and <i>R. oryzae</i> from <i>D. viscosa</i>	<i>A. niger</i> TU 62, <i>C. madrasense</i> AUMC14830, and <i>R. oryzae</i> AUMC14823	17 polyphenolic compounds from the different extracts of host plants and their endophytic fungi were analyzed using docking studies to determine their binding affinities to target proteins as antioxidants	The most potent anti-free radical isolates were identified as <i>A. niger</i> TU 62, <i>C. madrasense</i> AUMC14830, and <i>R. oryzae</i> AUMC14823 by utilizing the ITS region sequencing	[27]
<i>Antimicrobial protein 2 (MiAMP2)</i> from <i>Macadamia integrifolia</i>	Peptides EQVR, EQVK, AESE, EEDNK, EECK, and EEVE	Screen novel dipeptidyl peptidase IV (DPP-IV) inhibitory peptides in antimicrobial protein 2 (MiAMP2) in silico digestion using ExPASy Peptide Cutter	Six novel peptides, EQVR, EQVK, AESE, EEDNK, EECK, and EEVE were predicted to possess good DPP-IV inhibitory potentials	[11]
By-products Miiuy croaker fish (<i>Milichthys miiuy</i>)	Angiotensin I-converting enzyme (ACE) inhibitory peptides: DEGPE, EVGIQ, SHGEY, GPWGPA, GPFQTD, SPYGF and VIGPF	Molecular docking to illustrate that activity of SHGEY and SPYGF with the affinity of – 8.7 and – 9.7 kcal/mol mainly attributed to effectively combining with their active sites by hydrophobic interaction, electrostatic force and hydrogen bonding	Peptides SHGEY and SPYGF are health-promoting ingredients for functional products as a supplementary treatment for hypertension and cardiovascular diseases	[32]
Camel milk	Inhibition of cholesterol esterase peptides: WPMLQPKVM, CLSPLQMR, MYQQWKFL, and CLSPLOFR	Novel bioactive peptides from camel milk protein hydrolysates (CMPH) were identified and tested for inhibition of cholesterol esterase (CEase), and their possible binding mechanisms were elucidated by molecular docking	WPMLQPKVM, CLSPLQMR, MYQQWKFL, and CLSPLOFR peptides showed inhibition of cholesterol esterase (CEase), which was able to bind to the active site of the enzyme. All peptides formed hydrogen bonds and hydrophobic interactions with catalytic pockets of the active site of CEase	[33]
Edible cricket	Peptides YKPRP, PHGAP, and VGPPQ	Three novel peptides, YKPRP, PHGAP, and VGPPQ, were chosen for the molecular docking studies to identify the mechanism of their bioactivity	PHGAP and VGPPQ exhibited a higher degree of non-covalent interactions with the enzyme active site residues and binding energies comparable to captopril	[34]

The generation process can be iterated until optimized scaffolds meeting predefined classification metrics are obtained, which serve as selection criteria for design refinement. During assays, parameters such as tree topology, evolutionary distance matrices, mutation rates, and insertion/deletion probabilities are adjusted to simulate the evolution of offspring from a parent sequence [37]. Once the iterative process is complete, a library of related compounds can be constructed and screened using *in silico* techniques, such as virtual screening or molecular docking. Once the iterative scheme is finished, compounds with the highest predicted bioactivity properties are tested *in vitro* or *in vivo* to validate bioactivity or therapeutic functions [38].

Implementing simulated molecular evolution events requires specialized computational tools and trained personnel with expertise in bioinformatics, cheminformatics, and molecular modeling. The computational aspects rely on using evolutionary algorithms and simulation software, which may require training in phylogenetic analysis and molecular dynamic simulations. Additionally, laboratory personnel must be familiar with iterative synthesis and testing protocols. The time required for such studies depends on the complexity of the target molecules and the number of iterative cycles performed, typically ranging from weeks to several months. Resource-wise, high-performance computing infrastructure, molecular modeling software (e.g., ROSE, AutoDock, or Schrodinger Suite), and access to experimental validation facilities are essential [39].

For example, anticancer peptides have been designed using molecular evolution simulation. Neuhaus et al. [35] successfully identified a total of 51 new anticancer peptides, including two (named as 1.7.3.10, a GWYEIIKKIYKWLK sequence, and 2.5, a FHAWAKLLKGVGRFFKGIGRW sequence) which exhibited better activity (EC₅₀ 4.0 μ M) against the breast cancer-7 cell line. Likewise, Ruiz-Blanco et al. [40] applied evolutionary principles (tree topology, evolutionary distance, mutation rate, insertions, and deletions) to construct diversity-oriented peptide libraries aimed at obtaining peptide inhibitors of ATP synthase. Using the Random Model of Sequence Evolution algorithm (ROSE <https://bibiserv.cebitec.uni-bielefeld.de/rose>), they screened an initial library of 5428 peptides. From this set, only two peptides (P1 and P2, both with 40 amino acids) were selected based on multiple criteria, including redundancy, interaction with target enzyme, differential interaction probabilities with *E. coli* and human enzymes, charge, and structural properties.

Overall, molecular evolution strategies provide a robust framework for optimizing bioactive compounds. However, their successful application necessitates a multidisciplinary approach involving computational biology, molecular biology, protein engineering, and access to appropriate computational and laboratory resources. Strategies like molecular evolution hold significant potential in food engineering and technology, enabling the development of functional ingredients, enhanced nutritional profiles, and novel bioactive compounds for improved food quality and health benefits. Figure 1 provides a schematic representation for *in silico* analysis that allows bioactivity mechanisms of compounds from different sources.

3 Natural compounds bioactivity: an *in silico* approach

In this section we discuss the growing interest in bioactive peptides and secondary metabolites, particularly in their therapeutic potential, which is being explored through *in silico* analysis. These bioactive compounds, obtained from natural sources like plants, animals, and microorganisms, are valued for their low toxicity and diverse biological activities, such

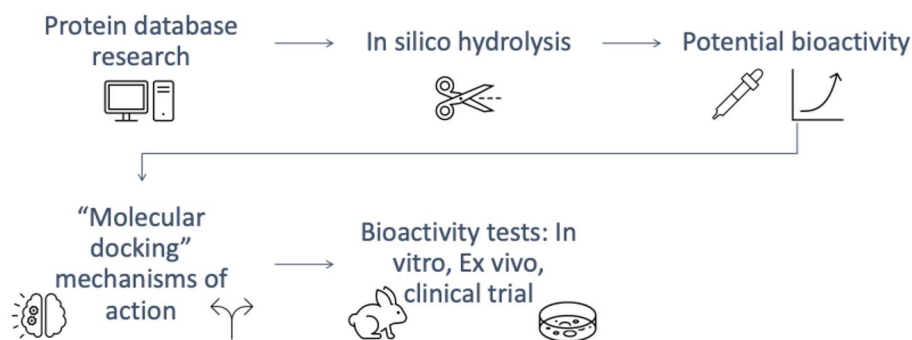


Fig.1 Steps in *in silico* assays to evaluate peptides from natural sources. Usually, similar steps are followed for *in silico* assays using secondary metabolites, except in *in silico* hydrolysis step. This integrative approach combines computational and experimental methods to identify and evaluate food-derived peptides with therapeutic potential

as antimicrobial, antihypertensive, anticancer, and antioxidant effects. Advances in bioinformatics enable the prediction of these compounds' bioactivities, structure–function relationships, and interactions with biological targets. *In silico* methods, including molecular docking simulations and enzyme inhibition modeling, provide insights into the efficacy of peptides and metabolites for addressing health conditions like oxidative stress, diabetes, and cancer. However, while promising, these findings require further experimental validation to confirm their practical applications.

3.1 Peptides and secondary metabolites used in silico analysis

During the last decades there has been an increase in research about peptides that can be used as a therapeutic alternative to their bioactive properties [41]. These bioactive peptides can be obtained by cooking, chemical hydrolysis, enzymatic digestion, fermentation, or direct isolation from foods or natural sources [42, 43]. The primary interest in these compounds as a natural therapeutic tool lies in their low toxicity, their capacity to bind to their targets with high affinity, and their different bioactivities reported for therapeutic peptides such as antimicrobial, antihypertensive, anticancer, antidiabetic, and others [40, 44, 45].

Plants and animals have traditionally served as a source of bioactive peptides [46]. Bovine milk, cheese, and dairy products are among the most significant food derived sources of bioactive proteins and peptides [47–49]. Bioactive peptides can also be obtained from other animal sources, such as bovine blood, gelatin, meat, eggs, fish, as well as from less explored non-animal sources, including seaweed, residues from the food industry, edible insects, cyanobacteria, and certain edible fungi. These alternative sources offer access to peptides with previously unexplored bioactivities [50, 51].

Primary metabolism requires essential compounds such as lipids, amino acids, carbohydrates, and nucleic acids necessary for cellular growth [52]. In contrast, secondary metabolites (such as alkaloids, flavonoids, phenols, terpenoids, and steroids) are low-molecular-weight molecules that are not directly involved in primary metabolism [53]. Secondary metabolites, mainly produced by plants, fungi, and bacteria, are therefore not essential for survival but exhibit wide range of interesting biological activities, including antibacterial, antifungal, antiviral, anti-inflammatory, and anticancer properties. Due to this properties, these metabolites are of great interest to pharmaceutical and agricultural industries. Many of these compounds are consumed as part of the diet, being naturally present in fruits, vegetables, cereals, beverages, fermented products, and mushrooms, where they contribute to sensory properties such as color and flavor [54].

Advances in bioinformatics, also known as *in silico* analysis, allow for the prediction and identification of peptides and other compounds with potential bioactivities. These computational techniques enable the elucidation of structure–function relationships and the proposal of mechanisms of action before experimental validation [55]. Databases such as BIOPEP, UniProtKB, NCBI, PepBank, focus primarily on food peptides embedded in the primary structure of the food proteins of interest [56]. Peptide cutter programs are used to generate *in silico* peptide profiles from specific primary protein structures using enzymes of known specificity [57, 58]. As previously mentioned, molecular docking simulations have also been developed to predict possible interactions with biological targets, e.g., active sites of enzymes [55]. *In silico* alternatives offer therefore practical, faster, and cost-effective analysis, when compared to *in vitro* or *in vivo* assays, providing important information before evaluating *in vitro* activities and further clinical trials (Fig. 2) [59].

Subsequent sections of this review describe different studies that primarily employ an *in silico* approach, sometimes complemented with *in vitro* assays, to optimize and analyze information about the bioactivity of different peptides and secondary metabolites. Natural compounds are categorized according to their reported bioactivity. For a clearer overview of the steps typically involved in *in silico* drug discovery from food-derived natural compounds, a visual summary is provided in the Fig. 3.

3.1.1 Antioxidant effect

Free radicals are one of the main factors contributing to oxidative stress, generated naturally in the body or through environmental contaminants [60]. While body synthesizes antioxidant molecules, many antioxidants come from dietary sources. Antioxidant molecules inhibit free radicals, protect the body against reactive oxygen species, and regenerate other dietary antioxidants [61]. Antioxidant properties, in turn, may help reduce oxidative stress, which is implicated in various health conditions. Numerous studies have investigated enzymes responsible for generating endogenous free radicals.

To enhance the value of by-products and identify effective antioxidant peptides from broken rice, *in silico* predictions have been used to identify and isolate 98 peptides, including four novel peptides with outstanding solubility and antioxidant activity (Table 2) [62]. Molecular docking simulations revealed that SGDWSDIGGR exerts its antioxidant effect

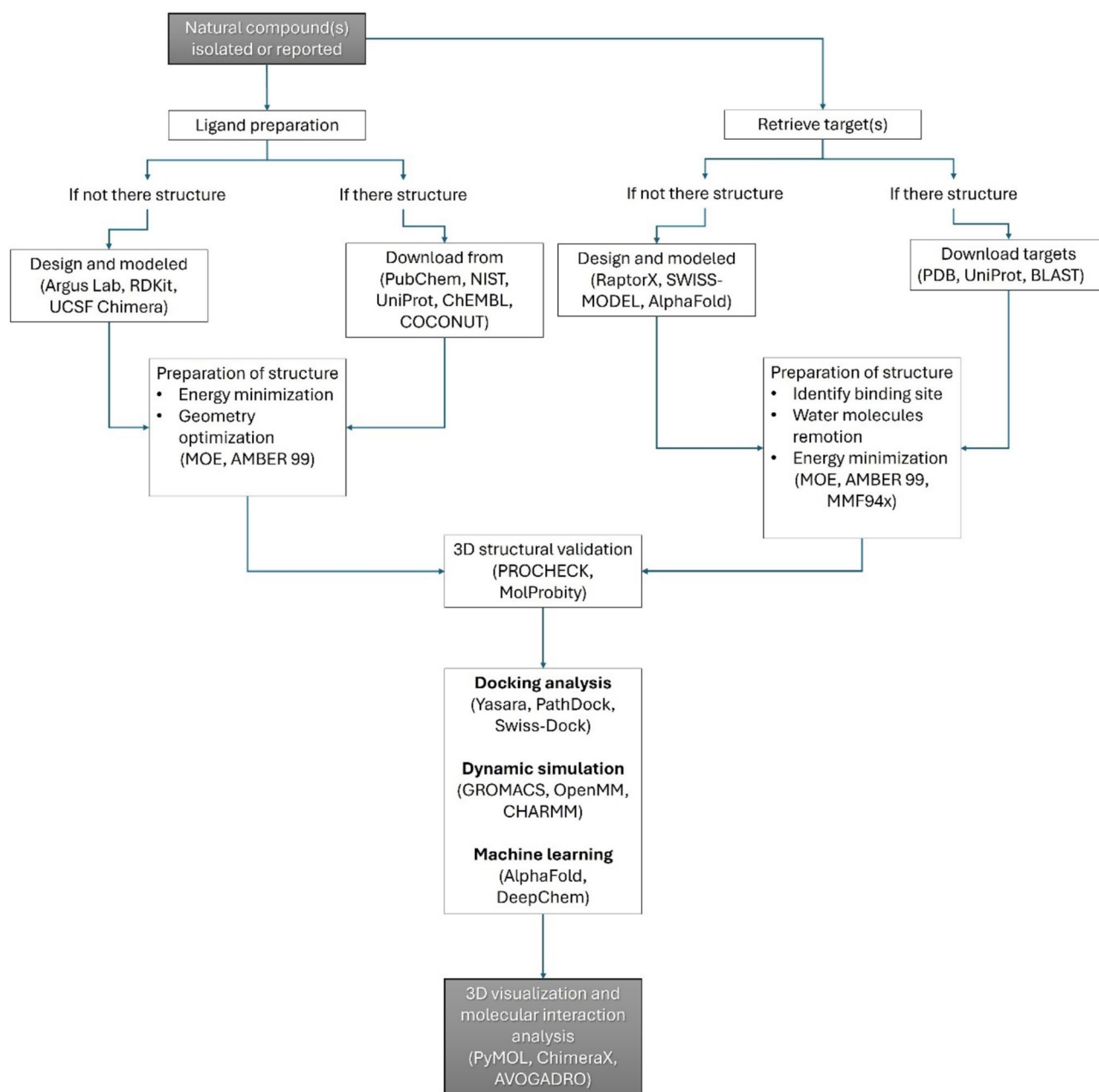


Fig. 2 In silico workflow for drug discovery from known or unknown compounds. In parenthesis, some servers or software are included. This flowchart provides a comprehensive overview of computational strategies used to identify, model, and evaluate natural compounds with therapeutic potential

by binding to the Kelch-like ECH-associated protein 1 (Keap1) at a binding energy of -103.22 kcal/mol. This interaction activates the Keap1-Nrf2 signaling pathway, upregulating antioxidant enzyme expression. Although the study discussed in [62] provides valuable insights into antioxidant peptide discovery, its findings rely solely on computational models, necessitating further in vitro and in vivo validation to confirm their physiological relevance and therapeutic potential.

Hassane et al. [27] evaluated the effect as antioxidant of polyphenols from black seeds (*Ilium ampeloprasum*, *Allium cepa*, *Amaranthus retroflexus*, *Dodonaea viscosa*, *N. sativa*, *Ocimum basilicum*, *Papaver somniferum*, *Piper nigrum*, and *Portulaca oleracea*) by assessing inhibition of myeloperoxidase, an enzyme involved in immune defense against infections responsible for reactive oxygen intermediates generation [63]. Of the seventeen compounds identified, rutin exhibited the highest affinity to the myeloperoxidase enzyme (Table 2). To achieve this result, a 3D modeling technique that includes three-dimensional potentiation energy minimization with Merck Molecular Forcefield (MMFF94x) was required.

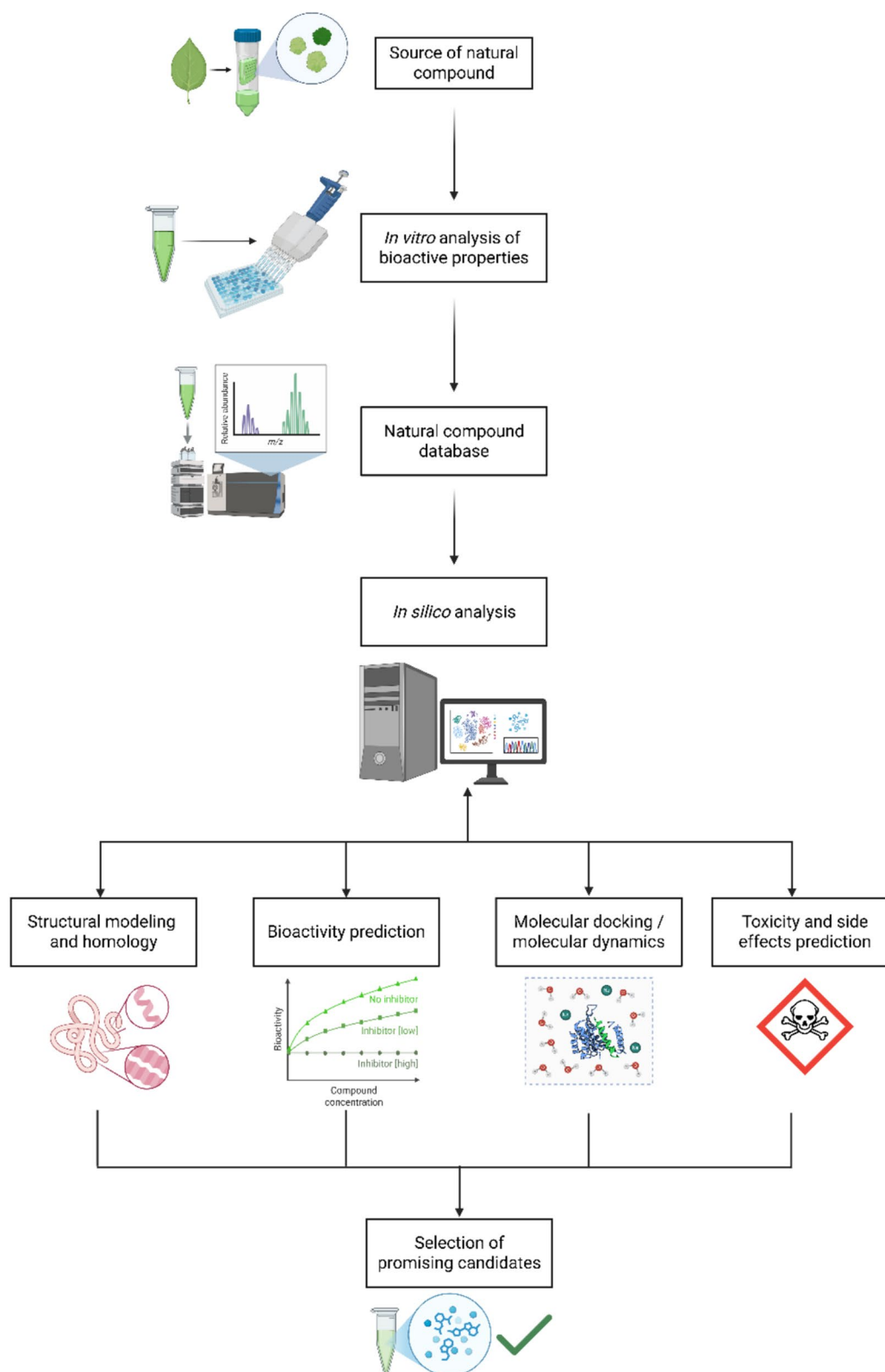


Fig. 3 Flowchart summarizing the key steps involved in the in silico pipeline for screening and optimizing food-derived natural compounds, including physicochemical analysis, bioactivity prediction, molecular modeling, and biological effect evaluation to identify promising drug candidates

Table 2 In Silico analysis of bioactive natural compounds with antioxidant, antidiabetic, and anticancer potential

Bioactivity	Target compound	In silico method	Outcome	Reference
Antioxidant effect	Polyphenols from black seeds	Molecular docking was performed using 3D crystal structure of myeloperoxidase (PDB ID: 1DNU)	Rutin exhibited the highest binding affinity (– 15.31 kcal/mol) to myeloperoxidase, indicating a strong potential as an antioxidant by preventing the release of reactive species	[27]
	Four peptides from broken rice, SGDWSIDIGGR, DFGSEILPR, GEPPSPDKKQLQ, GEKGIGIPIGK	Different tools were used to identify antioxidant peptides: BIOPEP, PeptideRanker, and ToxinPred. Molecular docking was performed using the 3D crystal structure of Keap1 (from PDB) to analyze peptide interactions	SGDWSIDIGGR showed strong activity (41.56 μmol TE/g) and protecting cells from oxidative stress. It activates the Keap1-Nrf2 pathway, enhancing antioxidant enzyme expression	[62]
	Polyphenolic compounds from <i>Cynarae folium</i> , <i>Rosmarini folium</i> , <i>Taraxaci herba</i> , <i>Cichorii herba</i> , and <i>Agrimoniae herba</i>	Molecular docking to evaluate the interactions between the identified polyphenols and key molecular targets related to liver disease, such as cytochrome P4502E1 (CYP2E1), tumor necrosis factor alpha (TNF-α), and glutathione peroxidase 4(GPx4)	<i>Agrimoniae herba</i> extract exhibited the highest antioxidant activity (IC ₅₀ ABTS 0.0147 mg/mL), whereas <i>Cynarae folium</i> extract showed the lowest activity (IC ₅₀ ABTS 0.1588 mg/mL). The study demonstrated a correlation between polyphenolic content and antioxidant potential, supporting their hepatoprotective effect	[64]
	Phytochemicals founded in <i>Carica papaya</i> , <i>Citrus limon</i> , <i>Curcuma longa</i> , <i>Dalbergia sissoo</i> , and <i>Punica granatum</i>	Molecular docking to evaluate agonist effect in enzymes involved in antioxidant pathway: Superoxide dismutase (SOD, PDB ID: 1CB4), Catalase (CAT, PDB ID: 2CAG), and Glutathione peroxidase (GPX, PDB ID: 2P31)	Extract from <i>Punica granatum</i> exhibited strong binding affinity with SOD and GPX (– 72.76 kcal/mol). Gliadin from <i>Triticum aestivum</i> showed the highest binding to SOD (– 111.03 kcal/mol), while Terrestrisamide from <i>Triticum aestivum</i> had the highest affinity to CAT (– 130.94 kcal/mol)	[65]
	LPLLR from walnut (<i>Juglans mandshurica</i> Maxim.)	In silico anti-diabetic analysis using BIOPEP database	At 2000 μM, LPLLR peptide inhibited α-glucosidase and α-amylase with inhibition rates reaching a maximum of 50.12% and 39.08, respectively. Moreover, it improved insulin resistance in HepG2 cells	[70]
	Peptides from <i>Ocimum basilicum</i> seeds (ACGNL-PRMC, ACNLPRMC, AGCGCEAMFAGA)	Molecular docking to evaluate the interactions between peptides and catalytic residue of α-amylase (PDB ID: 1PIG)	Peptides bound to substrate binding residues of α-amylase, restricting enzyme flexibility and preventing carbohydrate hydrolysis	[71]
	<i>Garcinia linii</i> extracts from leaves and stem	Molecular docking to evaluate the interactions between natural compounds extracted and different proteins templates from PDB (IDs: 3BLP, 5U3A, 2QJM, 3TOP, 4N8D, 4RER, and others)	<i>Garcinia linii</i> extracts exhibited significant inhibition of α-amylase (IC ₅₀ 13.24–17.24 mg/mL) and α-glucosidase (IC ₅₀ 0.02–0.05 mg/mL). Docking studies revealed higher binding affinity of <i>Garcinia linii</i> compounds compared to clinical drugs like Acarbose	[73]
	γ-Mangostin from <i>Garcinia mangostana</i>	Molecular docking using PubChem database for small molecules and PDB for proteins AMPK, DPP4, IR, and other enzymes involved in type-2 diabetes	γ-Mangostin inhibited α-amylase and α-glucosidase more effectively than Acarbose. Docking studies revealed strong binding affinities with AMPK, PPARγ, DPP4, and glucose metabolism-related enzymes, suggesting improved insulin sensitivity	[74]

Table 2 (continued)

Bioactivity	Target compound	In silico method	Outcome	Reference
Antidiabetic effect	Commercial syringaldehyde	Molecular docking using key proteins in diabetes disease. Structures from PDB were: GLP1 receptor (ID: 7C2E), DPP4 (ID: 4N8D), PPAR γ (4C15), Human acetylcholinesterase (ID:4MOE), Acetylcholine M2 receptor (ID: 5ZK8)	Syringaldehyde interacted strongly with GLP-1, PPAR γ , DPP-4, and acetylcholinesterase, suggesting a potential role in glucose homeostasis and diabetes treatment	[75]
	47 natural compounds antagonists against α -glucosidase and α -amylase	Molecular docking using 3D structure from PDB, α -glucosidase (ID: 2ZE0), and α -amylase (ID: 1HNY) enzymes	Docking analysis identified curcumin as the most potent inhibitor of both α -glucosidase and α -amylase, exhibiting stronger inhibition than acarbose (standard drug). Its binding affinity, measured by docking scores (-PMF, Potential of Mean Force), was highest at - 153 for α -glucosidase and - 175 for α -amylase, suggesting potential therapeutic applications in glucose regulation	[76]
	Polyphenols from red wine (resveratrol, catechin gallate, (-)-epicatechin 3-O-gallate, rosmarinic acid, piceatannol 3'-O-glucoside, apigenin, ellagic acid, δ -viniferin)	Molecular docking using crystal structures of AKT1 (ID: 3O96), STAT3 (ID: 6NJS), and IL-6 (ID: 1ALU) downloaded from PDB	Red wine polyphenols showed potential anti-diabetic effects. Thorough molecular docking analysis, δ -viniferin, resveratrol, and catechin gallate exhibited high binding affinities to key targets, including AKT1, STAT3, and IL6. These findings suggest a role in glucose metabolism regulation and protection against type 2 diabetes-related complications	[77]
	37 alkaloids reported as α -glucosidase inhibitors isolated from different food sources	Homology modeling and molecular docking	Oricacridone F (- 15.13 kcal/mol) and O-methylmanine (- 14.91 kcal/mol) exhibited the highest binding affinity, comparable to standard inhibitors miglitol (- 15.44) and acarbose (- 14.79). These compounds formed hydrogen bonds and hydrophobic interactions with key active site residues of α -glucosidase, suggesting strong inhibitory potential	[79]

Table 2 (continued)

Bioactivity	Target compound	In silico method	Outcome	Reference
Anticancer effect	Compounds isolated from <i>Vitex negundo</i> L	Molecular docking using the 3D structures of proteins from the PDB, 7KCD (Keap1) and 2G33 (myeloperoxidase) which are involved in oncogenesis and cancer progression	Artemetin, vitexicarpin, and penduletin showed strong anticancer activity. Molecular docking revealed their potential as inhibitors targeting specific cancer pathways	[83]
	Polyphenols from different parts of <i>Syzygium alternifolium</i>	Molecular docking using breast cancer estrogen receptor α (ER α) structure from PDB (ID: 1A52)	Molecular docking showed strong interactions with key residues of Era and 7 polyphenols, indicating potential anticancer activity	[95]
	Four anticancer biopeptides designed: Amphi-Arc2, Off2, Off2.2.10, Gradient2	Machine learning and simulated molecular evolution	AmphiArc2 showed micromolar activity against A549 and MCF7 cancer cells. Off2 improved selectivity and reduced hemolytic activity. Off2.2.10 improved selectivity while retaining anticancer activity. Gradient2 active against MCF7 cancer cells	[93]
	Alkaloids from <i>Zanthoxylum</i> (Rutaceae)	Machine learning using PASS (Prediction of Activity Spectra of Substances)	32 alkaloids demonstrated the ability to activate the apoptotic cascade and inhibit key oncogenic genes	[97]

After minimal energy conformers were selected, the docking procedure (Molecular Operating Environment software MOE 2019.01) was used to evaluate the binding patterns with enzymatic proteins (Protein Data Bank). Although the study demonstrated the antioxidant potential of polyphenols, further kinetic assays and enzymatic inhibition experiments are necessary to fully understand their biological effects, particularly their metabolic transformations *in vivo*, which may influence their binding efficacy and overall functionality.

Enzymes may be inhibited or activated by the presence of metabolites obtained from natural sources. Enzymes such as cytochrome P450 2E1 (CYP2E1), a key enzyme in metabolizing polyunsaturated fatty acids to toxic metabolites; the tumor necrosis factor- α (TNF- α), that causes inflammation, oxidative stress, and hepatocyte apoptosis; and the antioxidant enzyme glutathione peroxidase 4 (GPx4), that prevents hepatocellular degeneration by suppressing lipid peroxidation and inflammation, have been proved to interact with metabolites from *Cynarae folium*, *Rosmarini folium*, *Taraxaci herba*, *Cichorii herba*, and *Agrimoniae herba* (Table 2) [64]. Molecular docking has been used to show that pinocembrin shows the highest binding affinity to CYP2E1, rutin to TNF- α , and Naringin for GPx4. These findings should be interpreted with caution as the referenced study lacks experimental validation of these interactions and does not account for the tested compounds' potential off-target effects or bioavailability. Ligand efficiency analysis also suggested that these molecules exhibited the lowest ligand efficiency, raising concerns about their efficacy in physiological conditions.

In addition to polyphenols, other phytochemicals have also been investigated for their potential properties. Rana et al. [65] tested phytochemicals found in *Carica papaya*, *Citrus limon*, *Curcuma longa*, *Dalbergia sissoo*, and *Punica granatum*, as agonists in enzymes involved in the antioxidant pathway (reducing reactive oxygen species/free radicals). Superoxide dismutase was found to bond better with tea extract (phytochemicals present are not listed) from *P. granatum*, violaxanthin from *C. papaya*, hesperidin from *C. limon*, mutatochrome from *C. papaya*, glutathione peroxidase with tea extract (*P. granatum*), vitamin P (*C. papaya*), trigalloyglucose (*P. granatum*); catalase with terrestribisamide (*T. aestivum*), vitamin P (*C. papaya*), aurochrome (*C. papaya*) and curcuminoids (*C. longa*). Although using antioxidant enzymes as targets is an innovative approach for *in silico* studies, the study does not clarify whether these phytochemicals exhibit direct enzymatic modulation in biological systems.

Other enzymes that may be targeted as reactive oxygen species generators include lipoxygenase, NADPH oxidase, and xanthine oxidase [66]. Additionally, peroxiredoxin 5 (PDB ID: 1HD2), the anti-inflammatory protein Human Cyclooxygenase-2 (PDB ID: 5IKQ) [65, 67], and SPSB2 [68] have also been identified as potential targets and could be used for further studies.

3.1.2 Antidiabetic effect

Diabetic diseases result from disruptions in key cellular pathways, including insulin secretion, insulin resistance, and carbohydrate absorption. Several biomolecules play a role in these processes, such as glucokinase, AMP-activated protein kinase, 11 β -hydroxysteroid dehydrogenase, insulin receptor substrate, interleukin-1 beta, dipeptidyl peptidase IV (DPP-IV), C-reactive protein, glutamine fructose-6-phosphate aminotransferase, peroxisome proliferator-activated receptor gamma, α -amylase, α -glucosidase, protein tyrosine phosphatases, tyrosine kinase insulin receptor, protein kinase B, and the insulin receptor [69].

Research on DPP-IV inhibitory peptides has identified aromatic amino acids with polar side chains and proline at the N-terminus, including WP and YP dipeptides, as key structural features [58]. Similarly, the inhibition of α -glucosidase requires peptide sequences of three to six amino acid residues, featuring S, F, Y, K, or R at the N-terminus and a P residue near the C-terminal, with M or A at the C-terminal position [59]. Peptide LPLLR effectively inhibits both α -glucosidase and α -amylase [70]. Additionally, peptides derived from basil seeds, such as P1 (ACGNLPRMC), P2 (ACNLPRMC), and P3 (AGCGCEAMFAGA), were shown to bind to different residues in the catalytic site of α -amylase, as confirmed by *in silico* structural modeling (Table 2) [71, 72].

In vitro and *in silico* approaches are often used complementarily to identify and validate bioactive peptides within complex food matrices. Dietary polyphenols have been extensively studied for their potential role in prevention of diabetes. Multiple studies suggest that polyphenols inhibit α -amylase and α -glucosidase, thereby reducing carbohydrate hydrolysis and absorption (Table 2). Notable compounds that exhibit enzyme suppression through various mechanisms, elucidated via *in silico* analysis, include syringaldehyde from *Garcinia linii*, γ -mangostin, curcumin, 16-hydroxy-cleroda-3,13-diene-16,15-olide (16-H), docosanol, tetracosanol, quercetin, rutin, and caulerpin from *Caulerpa racemosa* algae [73–77]. However, many studies focus on isolated polyphenols rather than whole-food matrices, raising concerns about bioavailability, metabolism, and potential synergistic or antagonistic effects in real dietary contexts.

Diker and Kutluay highlighted the potential of complex natural sources in preventing diabetes due to their multi-component, multi-target effects [78]. For instance, polyphenols from black beans and blue corn were tested against 13 proteins involved in type 2 diabetes mellitus, with anthocyanins demonstrating strong binding properties [69]. Similarly, Zafar et al. examined 37 alkaloids against α -glucosidase using molecular docking techniques. Oriciacridone F and O-methylmahanine exhibited the lowest binding energy, interacting through hydrogen bonding, hydrophobic interactions, and arene-cation interactions. Tyrosinase and α -amylase were also evaluated as therapeutic targets, with the alkaloid pakistanine proving to be an effective inhibitor. Similarly, Zafar et al. examined 37 alkaloids against α -glucosidase using molecular docking techniques [79]. Oriciacridone F and O-methylmahanine exhibited the lowest binding energy, interacting through hydrogen bonding, hydrophobic interactions, and arene-cation interactions. Tyrosinase and α -amylase were also evaluated as therapeutic targets, with the alkaloid pakistanine proving to be an effective inhibitor other alkaloids were shown to be effective inhibitors of these enzymes [80].

An alternative approach was proposed by Eliwa et al. [81], who investigated the inhibition of protein tyrosine phosphatase 1B (PTP1B), an enzyme involved in insulin resistance and obesity through the dephosphorylation of insulin receptor tyrosine residues. Their study demonstrated that various forms of berberine, a natural alkaloid, exhibit strong binding affinity for PTP1B, with binding interactions influenced by chemical structure and hydrogen bonding with different protein residues.

Given the multifaceted nature of diabetes, it is crucial to study multiple molecular targets rather than focusing on one or two enzymes. While many experimental studies assess limited enzymatic interactions, *in silico* approaches provide broader insights by evaluating multiple enzyme templates, enhancing the reliability and applicability of the findings.

3.1.3 Anticancer effect

Cancer arises from genetic and metabolic alterations influenced by genetic, environmental, and dietary factors [82]. These changes lead to uncontrolled cell growth, sometimes resulting in the production of dysfunctional proteins that impair cellular damage repair [83, 84]. Standard treatments against cancer include chemotherapy, radiotherapy, and surgery, alongside emerging therapies such as hormonal therapy, immunotherapy, nanotechnology, and RNA therapy [85]. However, current treatments often cause significant side effects, including hematological, gastrointestinal, and neurological complications, as well as general physical discomfort.

Certain natural compounds exhibit anticancer properties by modulating cell cycle signaling, enhancing the removal of anticancer agents, regulating antioxidant enzyme activity, inducing apoptosis, inhibiting tumor growth and metastasis, and arresting cell proliferation [82, 86]. Research into food-derived bioactive peptides with anticancer potential has therefore increased in recent years. However, while some peptides have demonstrated cytotoxic effects against cancer cells, further studies are necessary to confirm their safety for normal cells. To date, most bioactive peptides with anticancer activity have only been assessed *in vitro*, with limited knowledge of their *in vivo* effects [87].

The mechanisms underlying the anticancer properties of bioactive peptides remain under investigation. Some peptides target cancer cells by interacting with the negatively charged lipid layer of the cancer cell membrane [88–92]. However, the structure–activity relationships governing peptide selectivity for cancerous over healthy cells are not yet fully understood. To address this, researchers have developed machine-learning models to design selective anticancer peptides. A computational approach was experimentally validated through the synthesis and testing of 12 computationally generated peptides (Table 2), demonstrating the feasibility of designing optimized anticancer peptides with reduced hemolytic liability and increased selectivity [93]. However, many of these studies rely on *in vitro* models, and the real impact on complex biological systems remains unclear.

Polyphenols have also been studied for their anticancer potential, with various cell lines proposed as targets for their activity [94]. While isolated polyphenols exhibit bioactivity, combining multiple polyphenolic related micronutrients may enhance their anticancer effectiveness due to synergistic interactions [86]. *In silico* studies suggest that polyphenols can induce protein modifications. For example, Yugandhar et al. [95] identified several compounds from *Syzygium alternifolium*, including naringenin, eriodictyol, ($\pm$)-taxifolin, (-)-epicatechin, formononetin, acacetin, and hesperetin, which showed the lowest binding energies with the breast cancer estrogen receptor (ER α), a critical therapeutic target against breast cancer (Table 2). Additionally, phenolic compounds such as artemetin, vitexicarpin, and penduletin from *Vitex negundo* L. demonstrated promising effects against human hepatocellular carcinoma and breast cancer cells *in vitro*, exhibiting cytotoxicity against cancerous cells [83]. Yoshioka et al. [96] compiled evidence highlighting polyphenols' role in modulating cancer cell pathways, reinforcing their potential as therapeutic agents.

Other compounds like alkaloids were also investigated. Deyá [97] tested 32 alkaloids commonly found in the *Zanthoxylum* genus using the free web platform PASS (Prediction of Activity Spectra for Substances <http://www.way2drug.com/PASSOnline/>). The study revealed that most tested alkaloids exhibited anticancer activity by inducing apoptosis, modulating gene expression, causing nuclear and cytoplasmic alterations, and promoting DNA fragmentation. While computational predictions indicate promising anticancer properties, experimental validation is essential to confirm their effectiveness, specificity, and safety in noncancerous cells.

Table 2 compiles *in silico* several studies on bioactive compounds from natural sources evaluated for antioxidant, antidiabetic, and anticancer effects. Molecular docking and dynamics simulations were employed to assess interactions with therapeutic targets. These studies highlight the utility of *in silico* approaches in identifying and validating natural compounds for therapeutic applications.

3.1.4 Neuroprotective effect

Neurodegenerative diseases involve the progressive degeneration of neurological cells, which may manifest as memory, cognition, coordination, and/or motor impairments. Conditions such as Alzheimer's disease (AD), Parkinson's disease (PD), and multiple sclerosis have been extensively studied as potential targets for phytochemical-based therapies due to their complex pathophysiology and the limited effectiveness of current treatments.

A major pathological hallmark of AD is the accumulation of β -amyloid ($A\beta$) plaques and dysregulated acetylcholinesterase (AChE) activity, both contributing to cognitive decline. Kareti and Pharm [98] identified seven bioactive compounds in *Carissa carandas* extract (1-heneicosanol, N-nonadecanol-1, cholesta-4,6-dien-3-ol (3β), di-n-octyl phthalate, 7,9-di-tert-butyl-1-oxaspiro(4,5)deca-6,9-diene-2,8-dione, 6-undecyl-5,6-dihydro-2H-pyran-2-one, and phenol, 2,4-di-tert-butyl-6-nitro) that exhibited potential inhibition of both $A\beta$ aggregation and AChE activity (Table 3). Similarly, Ali et al. [99] investigated green tea polyphenols (catechin, epigallocatechin, epicatechin-3-O-gallate, and epigallocatechin-3-gallate) for their inhibitory effects on AChE and butyrylcholinesterase (BChE), enzymes linked to short-term memory loss. *In silico* studies assessed binding energy, amino acid interactions, and inhibition constants, showing binding energy values between -14.45 and -9.75 kcal/mol. Molecular docking assays suggest that tea polyphenols inhibit both AChE and BChE, prolonging cholinergic neurotransmission.

Additional studies have explored the neuroprotective effects of polyphenols through multiple mechanisms. The inhibition of key neurodegenerative enzymes—including β -secretase, γ -secretase, and glycogen synthase kinase-3 β (GSK-3 β)—has been observed with betanin, myricetin, and folic acid from *Beta vulgaris* [100], tea polyphenols [101], and proanthocyanidins from sorghum [102, 103]. These compounds have demonstrated therapeutic potential in mitigating neurodegeneration (Table 3). Polyphenols have also been reported to modulate ion channels and G protein-coupled receptors (GPCRs), which regulate electrolyte balance, cell volume, and neurotransmission. Catechin and epicatechin were identified as the most active compounds targeting GPCRs and ion channels [104], reinforcing their role in neuroprotection. Further research has focused on specific receptors, such as the Nav1.7 sodium voltage-gated channel, implicated in neuropathic pain. Sonvane et al. [105] reported that baicalin, butrin, dihydromonospermoside, icariin, isocoreopsin, and isosaponarin effectively inhibited this channel, particularly targeting the sulfonamide site, suggesting their potential in pain management associated with neurodegeneration.

In the context of PD, polyphenols have shown promising neuroprotective properties. Coffee-derived polyphenols have been investigated *in silico* for their potential to protect dopaminergic neurons, either by inhibiting monoamine oxidase (MAO) [106] or by binding to GPCRs (GPR-40/43). *In silico* and *in vitro* studies demonstrated that GPR-40/43 activation led to the release of glucagon-like peptide-1 (GLP-1), a key neuroprotective factor. Additionally, *in vivo* experiments showed that oral administration of 50 mg/kg for 13 weeks increased GPR-40/43 mRNA expression and reduced oxidative stress [107]. Other sources of polyphenols with potential therapeutic effects in PD include olive oil-derived 3-hydroxytyrosol [108] and various plant-based polyphenols [109, 110], further supporting their role in neuroprotection.

Beyond polyphenols, alkaloids have also emerged as promising neuroprotective agents. The alkaloid glaziovine was tested against 15 neurodegenerative-related proteins—including AChE, VGF nerve growth factor, cyclin-dependent kinase, GSK-3 β , NMDA receptor, β -secretase, TNF- α , adenosine A2A receptor, α -synuclein, MAO-B, and c-Jun N-terminal kinase (JNK)—demonstrating strong binding affinities and interactions comparable to clinically approved drugs [111]. Similarly, harmine and harmaline were tested for their potential against PD, targeting dopamine D2 and D3 receptors [112], revealing promising dopaminergic modulation. Hussain et al. [113] compiled a comprehensive review of alkaloids with potential neuroprotective effects, identifying isoquinoline, indole, pyrroloindole, oxindole, piperidine, pyridine, aporphine, vinca, β -carboline, methylxanthene, lycopodium, and erythrine derivatives as candidates for treating Alzheimer's,

Table 3 In silico analysis of natural bioactive compounds with neuroprotective, cardioprotective, and antimicrobial potential

Bioactivity	Target compound	In silico method	Outcome	Reference
Neuroprotective effect	Phytoconstituents extracted from <i>Carissa carandas</i> leaves	Molecular docking using Alzheimer's disease target proteins from the PDB. A β protein (ID: 2LMN) and AChE (ID: 3LIJ)	The top docking scores indicated strong binding interactions, demonstrating that 48 compounds from <i>Carissa carandas</i> leaf extract have potential inhibitory activity against amyloid β (A β) fibrils and acetylcholinesterase (AChE)	[98]
	Phytochemicals isolated from <i>Beta vulgaris</i> L	Molecular docking using protein structure of AChE from PDB (ID: 4BDT). Molecular dynamics simulation	Phytochemicals from <i>Beta vulgaris</i> demonstrated strong inhibitory activity against AChE, a key target in Alzheimer's disease. Betanin, myricetin, and folic acid exhibited high binding affinities (– 22 to – 16 kcal/mol) compared to donepezil (– 17 kcal/mol). Molecular dynamics confirmed the stability of ligand–protein complexes, supporting their potential as therapeutic agents for cognitive disorders	[100]
	Four tea polyphenols: catechin, epicatechin, and (–)-epigallocatechin gallate	Molecular docking using 5 enzymes involved Alzheimer's disease. 3D structure enzymes were recovered from PDB (IDs: 1FKN, 5FN2, 1Q5K, 4EY6, 4BDS). 3D structures of polyphenols were downloaded from the NCBI PubChem	Epigallocatechin gallate, showed high docking scores against key Alzheimer's targets: β -secretase, γ -secretase, GSK-3 β , AChE, and BuChE, suggesting its role as a multi-target therapeutic for Alzheimer's disease	[101]
	Natural products reported as anti-Alzheimer's disease agents	Molecular docking and molecular dynamics simulation. 2BEG and 2MXU were 3D templates downloaded from PDB	Molecular docking and dynamics confirmed that all natural compounds (rosmarinic acid, melatonin, and o-vanillin, apigenin, and quercetin) have inhibitory potency and antiaggregating activity. Apigenin and quercetin showed strong binding affinities to A β fibrils	[103]
	Caffeine derived from different products	Molecular docking using 2 receptor proteins involved in Parkinson's disease (IDs from PDB: 5MZZ, 2V5Z). COCONUT database was used to obtain 3D caffeine structures. Molecular dynamics simulation analyzes the stability of the ligand–protein complexes, providing insights into their potential neuroprotective effects	Molecular docking revealed two caffeine-based natural products as potent inhibitors of MAO-B and AAZAR, with binding scores of – 10.1 and – 9.7 kcal/mol, respectively. Molecular dynamics simulations demonstrated strong stability within the active site of MAO-B, suggesting their potential as neuroprotective agents for Parkinson's disease	[106]
	Six polyphenols from coffee	Molecular docking using 3D structures of the proteins GPR-40 (PDB ID: 4PHU) and GPR-43, which are recognized as therapeutic targets in Parkinson's disease. Additionally, molecular dynamics simulations were conducted to analyze the stability and interactions of the ligand–protein complexes	Molecular docking showed CGA showed strong binding affinity with GPR-40 and GPR-43, activating GLP-1 secretion, which provides neuroprotection in Parkinson's disease. Molecular dynamics simulations confirmed stable interactions, which could reduce oxidative stress and decrease phosphorylated alpha-synuclein accumulation	[107]

Table 3 (continued)

Bioactivity	Target compound	In silico method	Outcome	Reference
Cardioprotective effect	Alkaloids from <i>Nardostachys jatamansi</i>	Molecular docking using 15 Alzheimer's disease target proteins from PDB. Molecular dynamics simulations for evaluating ligand–protein interaction	Actinidine and glaziovine exhibited strong binding interactions with 15 neurodegenerative disease-related proteins, including acetylcholinesterase (AChE), β -secretase, NMDA receptor (GluN1/GluN2B), monoamine oxidase B (MAO-B), and glycogen synthase kinase-3 β (GSK-3 β). Glaziovine demonstrated superior binding energy to NMDA receptors and AChE, suggesting their potential as neuroprotective agents	[111]
	17 ethanolic compounds extracted from <i>Tacca leontopetaloides</i>	Molecular docking using 3D structure of HMG–CoA reductase downloaded from PDB (ID: 2R4F)	Stigmasterol was identified in ethanolic extract and exhibited strong binding affinity (– 7.2 kcal/mol) with HMG–CoA reductase, suggesting its potential to inhibit cholesterol biosynthesis highlighting its potential as a natural antihypercholesterolemia agent	[117]
	Peptide derived from β -casein hydrolysate	Molecular docking using 3D structure of thrombin (ID: 2BVR) from PDB. Peptide hydrolysis simulation using BIOPEP server	FOSEEQQQTEDELQDK exhibited high binding affinity to thrombin (252.387 kcal/mol), comparable to known thrombin inhibitors such as hirudin, suggesting its potential as a novel food-derived antithrombotic peptide	[120]
	Peptides derived from cowpea β -vignin	Molecular docking using 3D structure of HMG–CoA reductase from PDB (ID: 1HW9). In silico hydrolysis of cowpea β -vignin sequence obtained from UniProtKB. Simulated digestion of the β -vignin protein performed on the BIOPEP server	Molecular docking studies indicated that the peptides have a higher affinity for the substrate binding site. Through in silico and in vitro experiments was observed that IAF, QGF and QDF are capable of inhibiting HMG–CoAR activity via statin-like mechanism, reducing cholesterol synthesis	[124]
	Peptides from Pacific cod (<i>Gadus macrocephalus</i>) skin	Molecular docking using ACE metalloprotease (PDB ID: 1O8A)	GASSGMGP and LAYA peptides showed strong binding to ACE with a free energy of binding of – 5.16 kcal/mol and – 4.88 kcal/mol, respectively. Both peptides showed significant ACE inhibitory effects, making them potentially functional for cardiovascular disease prevention	[127]
Antimicrobial effect	Flavonoids and phenolic acids identified in commercial beers	Molecular docking using 3D structures of serum proteins (PDB IDs: 1B09, 1H9Z, 2R37, 3GHG)	Flavonoids and phenolic compounds from beer interact with key human serum proteins, enhancing antioxidant activity and providing cardiovascular protection. They exhibit a strong interaction with fibrinogen (3GHG), suggesting potential anticoagulant activity	[129]
	Flavonoids, non-flavonoids, and phenolic acids compounds in commercial red wine	Molecular docking using 3D structures of serum proteins (PDB IDs: 1B09, 1H9Z, 2R37, 3GHG)	Red wine polyphenols interact with key human serum proteins, enhancing antioxidant properties and potentially providing cardiovascular protection. Strong binding affinity to fibrinogen (– 7.9 kcal/mol for rutin, – 6.4 kcal/mol for tannic acid), suggesting anticoagulant activity	[130]
	Antimicrobial peptides (AMPs) from rapeseed	In silico enzymatic hydrolysis using BIOPEP database. Different tools based on Machine learning to predict the production of AMPs	Identified 26 novel AMPs; trypsin was the most effective enzyme for AMP production; Cruciferin generated more AMPs than Napin and Oleosin; peptides exhibited non-toxic and non-allergenic properties	[133]

Table 3 (continued)

Bioactivity	Target compound	In silico method	Outcome	Reference
	Ethanolic polyphenols extracted from eight edible plants	Molecular docking using 3D structures of fungal enzymes downloaded from PDB (IDs: 5FRB and 6K3H). Swiss Prot and Pubchem to obtain 3D structure of ligands	Rutin, kaempferol and quercetin founded in the ethanolic extracts, showed to inhibit fungal strains via inhibition of 14-alpha demethylase (CYP51) and nucleoside diphosphokinase (NDK), same as that of azole drugs	[135]
	Rosmarinic acid extract from <i>Perilla frutescens</i> leaves	Molecular docking using 3D structure of rosmarinic acid (ligand) obtained from PubChem database. 3D structure of proteins from PDB, Bacterial Peptide Deformylase (ID: 1LRU) and N-myristoyltransferase (ID: 1IYL)	Molecular docking revealed that rosmarinic acid strongly binds to bacterial peptide deformylase (– 7.9 kcal/mol) and N-myristoyltransferase (– 7.1 kcal/mol), effectively inhibiting key bacterial and fungal enzymes. These findings suggest that rosmarinic acid has potential as a broad-spectrum antimicrobial agent	[137]
	11 polyphenols from plants	Molecular docking using 3D structure of polyphenols from PubChem database. 3D structure of protein FabH downloaded from PDB, (ID: 5BNR)	Genistein exhibited the highest binding affinity to FabH (– 8.6 kcal/mol), followed by 4-naphthoquinone (– 8.579 kcal/mol) and pelargonidin (– 7.651 kcal/mol). These compounds demonstrate potential inhibition of bacterial fatty acid biosynthesis, suggesting that polyphenols are promising agents against bacterial infections	[140]
	Polyphenolic compounds from <i>Prunus persica L</i>	Molecular docking was performed using the 3D MOE builder tool and the 3D structure of the LasR protein downloaded from PDB (ID: 2UV0)	Molecular docking showed strong interactions with transcriptional regulator LasR. Quercetin, chlorogenic acid, and gallic acid exhibited the highest binding affinity to inside the active site of transcriptional regulator LasR (– 8.8, 6.1, and 5.8 kJ/mol, respectively). Therefore, the antimicrobial and antibiofilm effects of <i>Prunus persica L.</i> may be attributed to its polyphenolic compounds	[141]
	Polyphenols extract from <i>Cystoseira trinodis</i> , <i>Padina boryana</i> , and <i>Turbinaria triquetra</i> seaweeds	Molecular docking using PBP6 protein from <i>E. coli</i> (PDB ID: 3ITA), MurB protein from <i>S. aureus</i> (PDB ID: 1HSK), and SAP5 protein from <i>C. albicans</i> (PDB ID: 2QZX)	Docking assays revealed that 1,2-Benzenedicarboxylic acid exhibited the highest binding affinity among the tested compounds, showing strong interactions with the PBP6, MurB, and SAP5 proteins. The binding scores were – 26.3 kcal/mol, – 14.4 kcal/mol, and – 14.8 kcal/mol, respectively	[142]
	Chitosan dialdehyde (ChDA)	Molecular docking using a 3D membrane protein from <i>E. coli</i> (PDB ID: 7B53) and fungal peptide from <i>Candida albicans</i> (PDB ID: 1IYL)	ChDA exhibited significant antibacterial and antifungal activity by interacting in the binding pockets of 7B53 and 1IYL proteins. It demonstrated strong interactions with the bacterial cell wall and fungal peptide, highlighting its potential as a promising bioicidal compound with antibacterial and antifungal properties	[143]
	Alginate film loaded with acetyl-11-keto-β-boswellic acid (AKBA) from <i>Boswellia sacra</i> and silver nanoparticles	Molecular docking using molecular structure of AKBA downloaded from PubChem and sodium alginate from MOE database	Molecular docking analysis showed that AKBA molecules interact through hydrogen bonds, while silver interacts through ionic bonds. The composite film of sodium alginate, AKBA, and silver atoms complex showed a docking score between – 5.01 and – 4.63 kcal/mol. Indicating stability of the composite, which could improve antifungal properties	[144]

Table 3 (continued)

Bioactivity	Target compound	In silico method	Outcome	Reference
	Isoquinoline alkaloids of <i>Macleaya cordata</i>	Molecular docking and virtual screening using PharmaDB, HypoDB, and MOE builder tool	Molecular docking revealed that 6-acetyl-dihydrosanguinarine, chelerythrine, and sanguinarine exhibit inhibition of bacterial growth. This is due to their interaction with key targets, including transcriptional regulators, cell division proteins, and key enzymes in fatty acid biosynthesis, essential for bacterial growth and regulation, making these alkaloids promising candidates for antibacterial applications	[145]
	Phenolic compounds from Yerba mate (<i>Ilex paraguariensis</i> A.St. Hil) and jarilla (<i>Larrea divaricata</i> Cav.)	Molecular docking using a 3D model of main protease (Mpro) of SARS-CoV-2 virus	Quercetin-3-O-rutinoside from <i>Ilex paraguariensis</i> and 3,4-Dicaffeoylquinic acid from <i>Larrea divaricata</i> showed higher binding affinity (-9.6 and -7.6 kcal/mol) to Mpro protease, suggesting that extracts of yerba mate and jarilla could enhance defenses against SARS-CoV-2 virus	[149]
	Polyphenols from <i>Alchemilla viridiflora</i> Rothm	Molecular docking using a 3D model of S-glycoprotein of SARS-CoV-2 virus (PDB ID: 7BZ5) and neuropilin 1 (PDB ID: 2QQI)	Quercetin and pentagalloylglucose were two compounds with the higher binding energies of -8.035 (S-glycoprotein) and -7.685 kcal/mol (NRP1), respectively. The polyphenols found could have potential synergistic activity against SARS-CoV-2 virus	[151]

DPP4: Dipeptidyl peptidase 4; IR: Insulin receptor kinase domain; AMPK: AMP-activated protein kinase. AKT1: serine/threonine kinase STAT3: signal transducer and activator of transcription 3, IL-6: Interleukin 6

Huntington's, Parkinson's, epilepsy, schizophrenia, and stroke. This review serves as a valuable resource for future in silico investigations into alkaloid-based therapies for neurodegenerative disorders.

Overall, the expanding research on phytochemicals as neuroprotective agents highlights their multifaceted mechanisms of action against neurodegeneration. Computational approaches such as molecular docking, molecular dynamics simulations, and virtual screening have significantly contributed to identifying lead compounds with therapeutic potential. However, while in silico findings are promising, further in vitro and in vivo validation is essential to confirm efficacy, safety, and pharmacokinetic properties before clinical applications can be considered.

3.1.5 Cardiovascular effect

Cardiovascular diseases (CVDs) are among the leading causes of mortality worldwide, with the World Health Organization (WHO) estimating that 32% of global deaths are attributed to this disorder [114]. CVDs encompass a range of conditions affecting the heart and blood vessels, including coronary artery disease, cerebrovascular disease, venous thromboembolism, and peripheral vascular disease, which can lead to myocardial infarction, cardiac arrhythmias, or stroke [115]. The main risk factors for CVDs include poor diet, lack of physical activity, and the consumption of harmful substances such as tobacco and alcohol. Although these disorders are life-threatening, in silico studies have demonstrated that natural compounds can interact with key enzymes in blood serum [116] and the liver [117], offering potential therapeutic applications (Table 3).

Blood coagulation is a natural process that changes circulating substances within the blood into an insoluble gel, essential for survival. This process is carried out in abnormal vascular conditions or the absence of endothelial surface in the case of vascular injury [118]. However, excessive blood coagulation and clot formation can lead to severe medical conditions, necessitating the use of antithrombotic agents, particularly natural ones. Bioactive peptides have been identified as effective antithrombotic agents, functioning as both anticoagulants and antiplatelets. Anticoagulants prevent clot formation and growth, while antiplatelets inhibit platelet aggregation [119].

Tu et al. [120] used in silico screening to identify potential antithrombotic peptides from casein. The study identified SLVDAIGMGP and AGFAGDDAPR; these peptides were found to interact with thrombin exosite 1, a site on the thrombin protein that is involved in clotting. Non-common sources, such as insects, have been found to contain antithrombotic peptides, SLVDAIGMGP and AGFAGDDAPR have been identified in *Tenebrio molitor*, commonly known as mealworm beetle (Table 3). These potentially new antithrombotic agent sequences were not found in the reported *T. molitor* protein or peptide by Swiss-prot peptide search (<https://www.uniprot.org/peptidesearch/>). These peptides contained a PR fragment at the C-terminus, which is known to be part of antithrombotic peptides such as GPRG, GPRGP, GPRGPA, GPRGPP, and GPRPP, collagen-related synthetic peptides. [121, 122].

The search for bioactive peptides with cholesterol-lowering effects has intensified due to the adverse side effects of synthetic cholesterol-lowering drugs, which may lead to liver damage, myopathy, and diabetes in some individuals. Elevated cholesterol levels contribute to arteriosclerosis, a condition that restricts oxygen supply to the heart and increases the risk of CVDs [123]. Using in silico and in vitro approaches, Silva et al. [124] identified IAF, QGF, and QDF peptides from cowpea that function as HMG-CoA reductase (HMG-CoA) inhibitors via a competitive statin-like mechanism. These peptides significantly reduced cholesterol levels, as confirmed by in vitro inhibition studies, demonstrating inhibition rates of 69%, 77%, and 78%, respectively (Table 3). Similar research has reported that cumin seed-derived peptides (CSP1, CSP2, CSP3) can inhibit cholesterol micelle formation, lipase activity, and bile acid binding, suggesting that their consumption may lower cholesterol levels [125].

Additionally, polyphenols have been studied for their ability to regulate lipoprotein lipase, an enzyme implicated in cholesterol metabolism and lipid accumulation in arteries. In silico and in vivo studies have demonstrated that quercetin, rutin, and naringenin reduce lipoprotein lipase activity. However, their efficacy varies depending on their chemical structure (e.g., glycosides, aglycones, chalcones) and intestinal microbiota metabolism [116]. Another key enzyme in cholesterol biosynthesis, 3-hydroxy-3-methylglutaryl-coenzyme A reductase (HMG-CoAR), has been shown to be inhibited by stigmasterol through van der Waals forces and hydrophobic interactions. In these simulations, simvastatin (a conventional cholesterol-lowering drug) was used as a control, and results indicated that both compounds effectively inhibited HMG-CoAR activity and interacted with different amino acid residues [117].

Food-derived antihypertensive biopeptides have gained significant attention due to their high tissue affinity and prolonged bioavailability compared to synthetic drugs [126]. Several peptides have demonstrated angiotensin-converting enzyme (ACE) inhibition, a well-known mechanism for reducing hypertension and cardiovascular risk. Two peptides, GASSGMGP and LAYA, were isolated from Pacific cod (*G. macrocephalus*) skin gelatin using pepsin hydrolysis and

molecular docking (Table 3), confirming their strong ACE inhibitory potential [127]. Similar findings have been reported for IVDR, WYK, and VSAVI peptides from olive flounder (*P. olivaceus*) surimi, as well as LWHTH from tunicate (*S. clava*). In silico simulations demonstrated that these peptides bind to the active site of ACE, forming a stable ACE-LWHTH complex. Moreover, in vivo studies indicated that LWHTH significantly reduced blood pressure in hypertensive rats [128].

Regularly consumed beverages, such as beer and wine, are rich sources of natural phytochemicals that have been shown to interact with serum proteins and reduce cardiovascular risk (Table 3). Beer contains phenolic acids and flavonoids, which interact with human serum albumin and plasma fibrinogen, potentially contributing to cardioprotective effects. Studies suggest these compounds may also interact with C-reactive protein and glutathione peroxidase, which are involved in oxidative stress regulation [129]. Similarly, red wine polyphenols, particularly rutin, and resveratrol, have been shown to form covalent bonds and hydrogen interactions with key amino acids in C-reactive proteins. Additionally, rutin interacts with fibrinogen, human glutathione peroxidase 3, and human serum albumin, supporting its potential role in reducing inflammation and thrombosis risk [130].

Bioactive peptides and polyphenols exhibit promising cardiovascular benefits, ranging from antithrombotic and cholesterol-lowering effects to hypertension control and cardioprotective interactions with serum proteins. In silico models have been instrumental in identifying these compounds and predicting their binding affinities and mechanisms of action, paving the way for further validation through in vitro and in vivo studies. Continued research in computational modeling and functional food development could enhance the understanding and therapeutic application of these bioactive compounds in cardiovascular disease prevention and management.

3.1.6 Antimicrobial effect

The increase in antibiotic resistance has become a serious global health problem, compromising the effectiveness of conventional antimicrobial treatments. Given this situation, researchers have focused on alternative antimicrobial agents, such as antimicrobial peptides (AMPs), polyphenols, and alkaloids, demonstrating broad-spectrum activity against bacteria, fungi, and viruses. These properties make them better alternatives for conventional antibiotics, which have recorded resistance among pathogenic bacteria. Furthermore, antimicrobial peptides can function differently, directly killing the bacteria by making pores through the bacteria cell membrane or interacting with macromolecules inside the microbial cells [131, 132].

Antimicrobial peptides are particularly promising because they disrupt bacterial membranes or interact with key intracellular targets. Many AMPs are rich in positively charged amino acids, such as arginine and lysine, which facilitate their entry into microbial cells via energy-dependent endocytic pathways, including macropinocytosis. Duan et al. [133] used in silico screening to identify six antimicrobial peptides from *Brassica napus* (rapeseed), characterized by a high content of glycine (G) and glutamine (Q) residues. These findings align with previous studies that show that certain amino acid residues are preferred at specific positions of antimicrobial peptides. For instance, G, F, V, and R are abundant at the N-terminal position, while K, G, C, and R are commonly at the C-terminal position [134].

Beyond peptides, polyphenols have been identified as effective antimicrobial agents due to their ability to interact with essential microbial proteins and enzymes. While polyphenols are typically associated with antioxidant effects, certain structural configurations allow them to act as pro-oxidants, conferring antimicrobial, antifungal, and antipathogenic properties, depending on their chemical composition [135, 136]. The antimicrobial action of polyphenols has been linked to the inhibition of key bacterial and fungal enzymes. Li et al. [137] identified rosmarinic acid-rich extracts from *Perilla frutescens* with strong activity against *Escherichia coli*, *Staphylococcus aureus*, and *Candida albicans* (Table 3). The antimicrobial effect was attributed to the inhibition of peptide deformylase (in bacteria) and N-myristoyltransferase (in fungi), both previously proposed as drug targets for bacterial and fungal infections [138, 139]. Other polyphenols, such as rutin, kaempferol, and quercetin, have shown strong binding affinity to fungal 14- α demethylase and nucleoside diphosphokinase, enzymes critical for fungal survival. This interaction suggests these compounds could effectively inhibit mold growth [135].

Polyphenols have also demonstrated significant bactericidal effects by targeting β -ketoacyl-acyl carrier protein synthase III (FabH), an enzyme essential for bacterial fatty acid biosynthesis (Table 3). Molecular docking studies have shown that genistein, 4-naphthoquinone, and pelargonidin effectively bind to this enzyme, inhibiting bacterial growth [140]. Another mechanism of action involves disrupting bacterial biofilm formation, a key factor in antibiotic resistance. Rafey et al. [141] reported that polyphenols gallic acid and ferulic acid, derived from *Prunus persica* L., inhibited biofilm formation by targeting the transcriptional regulator LasR, a major component in bacterial quorum sensing. Osman et al. [142] reported that polyphenols extracted from three marine seaweeds (*Cystoseira trinodis*, *Padina boryana*, and *Turbinaria*

triquetra) exhibit antimicrobial activity against *E. coli*, *P. vulgaris*, *C. albicans*, and *A. fumigatus*, with a particularly strong effect against Gram-positive bacteria such as *B. subtilis* and *S. aureus*. The study suggests these isolated compounds inhibit cell wall synthesis by disrupting peptidoglycan remodeling or ergosterol biosynthesis. Additionally, they may alter membrane integrity, induce oxidative stress, and interfere with key metabolic enzymes, contributing to their antimicrobial effects (Table 3).

Some polysaccharides, which have valuable packaging formulation and design properties, have also demonstrated antimicrobial effects. *In silico* assays have provided insights into their potential mechanisms of action. The chitosan dialdehyde biopolymer (ChDA) exhibited biocidal activity against bacterial and fungal growth, suggesting its potential for food packaging design and development [143]. Similarly, Jawad et al. [144] developed films formulated with sodium alginate, acetyl-11-keto- β -boswellic acid (AKBA), and silver nanoparticles. *In vitro* assays revealed that the film enhanced its antifungal activity against *C. albicans* compared to its pure components. Additionally, *in silico* assays indicated that the interaction between the material components (sodium alginate-AKBA-Ag nanoparticles) has high binding scores, which could further enhance its antifungal properties (Table 3).

Alkaloids have also gained attention for their potent antimicrobial effects. Lei et al. [145] conducted a comprehensive screening of alkaloids from *Macleaya cordata* and tested their activity against *Pseudomonas aeruginosa*, *Escherichia coli*, *Bacillus subtilis*, *Tetracoccus* spp., and *Staphylococcus aureus*. Using DRUGBANK (<https://go.drugbank.com/>) and the Therapeutic Target Database (<https://db.idrblab.net/ttd/>), the study identified multiple potential targets involved in multidrug resistance, cell division, and fatty acid biosynthesis. While these findings suggest broad-spectrum activity, the study also noted that some proposed targets require further pharmacological validation.

In addition to bacterial and fungal infections, natural compounds have been investigated for their potential against viral pathogens, including SARS-CoV-2 (COVID-19). Several natural molecules have been shown to interfere with viral entry, replication, release, and diffusion [146, 147]. Computational studies identified thalimonine and shopaline D, two alkaloids with strong binding affinity to the SARS-CoV-2 main protease, suggesting potential antiviral activity [148]. Other *in silico* studies have explored plant-derived compounds from Yerba mate (*Ilex paraguariensis*), jarilla (*Larrea divaricata*), and *Alchemilla viridiflora*, among others, demonstrating potential inhibition of viral replication (Table 3) [149–151].

Table 3 presents *in silico* investigations into the therapeutic potential of natural compounds, utilizing molecular docking and dynamics simulations. Phytochemicals from various sources demonstrated neuroprotective effects, exhibiting affinity for targets associated with Alzheimer's and Parkinson's diseases. Cardioprotective compounds were found to inhibit HMG-CoA reductase and ACE, indicating their potential for cholesterol reduction and antithrombotic activities. Antimicrobial research highlighted the efficacy of polyphenols and algal extracts against bacterial and fungal enzymes, while alkaloids showed potential against SARS-CoV-2 through protease inhibition. These computational results highlight the wide-ranging bioactive potential of natural sources and advocate for further experimental validation.

3.2 Limitations and challenges of *in silico* approaches in bioactive compound discovery

In silico methods, such as molecular docking and dynamics simulations, serve as powerful tools for predicting the bioactivity of natural compounds. However, these computational techniques have limitations. A significant challenge concerns the potential inaccuracies of the models, which are deeply dependent on the quality of data input. This includes protein structures, which often rely on static crystal structures that may not accurately represent dynamic physiological conditions, as well as ligand parametrization. For example, scoring functions employed in docking can show biases towards specific chemical scaffolds and may neglect solvation effects, resulting in false positives or negatives. Furthermore, many algorithms prioritize binding affinity over pharmacokinetic properties, such as absorption, distribution, metabolism, and excretion (ADME), which are essential for drug development.

Another significant hurdle is translating computational predictions into experimental success. While a compound may exhibit strong binding *in silico*, its efficacy *in vitro* or *in vivo* can be affected by factors beyond the scope of simulations, such as cellular uptake, off-target effects, or metabolic instability. For example, a molecule with excellent docking scores might fail in cell assays due to poor membrane permeability or toxicity. Furthermore, the biological complexity of diseases, such as protein–protein interactions in neurodegenerative disorders or host-microbe dynamics in infections, often exceed the simplified systems modeled computationally.

These limitations highlight the need for thorough experimental validation. Both *in vitro* assays and *in vivo* studies are essential for verifying computational predictions and evaluating therapeutic potential. Collaborative processes that continuously refine *in silico* models using experimental feedback can improve predictive accuracy. Future developments may address gaps, including integrating machine learning with multi-omics data. Nevertheless, the synergy between

computational methods and wet lab research remains the gold standard for translating the discovery of bioactive compounds into real-world applications.

4 Conclusion

Different tools reviewed highlighted the effectiveness of following an *in silico* approach to designing and optimizing bioactive compounds from food sources. The strategies presented could be a way for rational design to produce new natural bioactive products. Although many lack *in vivo* models or clinical trials, their contributions could be considered for future research. The diverse bioinformatic tools have served to understand different mechanisms for bioactivity from natural compounds. These tools have emerged for the accurate design of peptides, secondary metabolites, and so on that can be used as therapeutic agents by enabling researchers to efficiently and cost-effectively identify promising lead compounds. The growth of these tools, together with continuous efforts by the scientific community, will make it possible to develop more robust bioinformatic tools that allow confronting some health problems such as degenerative diseases, cancer, and infections such as COVID-19. To advance this field, it is recommended that emerging trends such as deep learning and multi-omics analysis be integrated to improve predictive accuracy and that accessible computational platforms allow researchers in less-resourced regions to participate in drug discovery. However, these computational findings must be complemented with experimental validation to ensure their therapeutic relevance. The combination of advanced computational tools, global accessibility, and multidisciplinary collaboration could accelerate the development of innovative natural bioactive compounds.

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